

Ingegneria del Software

Design Thinking ed Elaborazione del progetto di Esame (&Challenge)

Prof. Paolo Giorgini

A.A. 2025-2026

IDEA Progetto

Design Thinking ; Requisiti ; Kanban ; Pitch ; User Stories;
Use Case Diagram
Sequence + Activity Diagram

D1

Implementazione

Git / GitHub
RESTful API
OpenAPI
Web 2.0 JavaScript
WebAPI Node.js
MongoDB
Authentication JWT + GoogleAuth
Frontend
Deployement & CI-CD
Testing Jest

D2

Analisi e Progettazione

Processi di sviluppo
Agile
Linguaggi di modellazione
Architetture
Component Diagram
Class Diagram
Testing

D3

Report finale

D4

4 weeks

10 weeks

lunedì 15 settembre	Introduzione al corso
martedì 16 settembre	Presentazione Comune di Trento
mercoledì 17 settembre	Ingegneria del software modelli e processi
venerdì 19 settembre	Design Thinking
lunedì 22 settembre	Design Thinking WP 1
martedì 23 settembre	Design Thinking WP 2
mercoledì 24 settembre	Linguaggi di modellazione
venerdì 26 settembre	Requisiti
lunedì 29 settembre	Presentazione documento D1
martedì 30 settembre	Agile
mercoledì 1 ottobre	Seminario: Kanban
venerdì 3 ottobre	Seminario su Agile: azienda Fincons Group

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Obiettivo del progetto



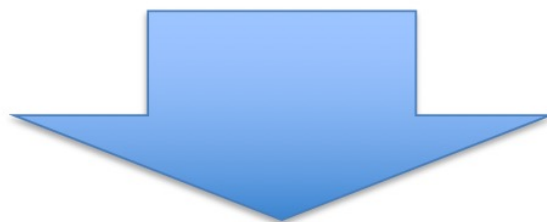
Smart City e Sostenibilità

Osservazioni:

- Il tema riguarda la **città di Trento**, non l'ente **Comune**
 - Il Comune è stato coinvolto in quanto è l'istituzione, ovvero l'ente che può rappresentare la città di Trento dal punto vista amministrativo.

Dall'esigenza reale al servizio per la città

Sulla base della vostra esperienza nella città di Trento, la sfida è nell'intercettare un'**esigenza** reale e da lì elaborare* un'**idea** che possa rispondere a tale esigenza



La vostra **idea** deve necessariamente sottointendere il concetto di **servizio**

* il ruolo del DT può essere funzionale per tale scopo

Concetto di servizio

- Nello sviluppo della vostra idea, pensate a quali **esigenze** il vostro servizio potrebbe rispondere, reiterate il ragionamento identificandone e complementando man mano con altre di rilevanza minore
 - Questo permetterà di dare consistenza al vostro servizio
- Integrate **input ed idee anche esterne al gruppo**, applicate metodologia di DT a livello di processo
 - Studiate il processo di DT, non datelo per scontato, rivedetelo durante lo sviluppo dell'idea.
- Ad alto livello, pensate a quali **informazioni** il vostro servizio potrebbe dover gestire per raggiungere lo scopo
 - Non dovete progettare un DB, è solo per ampliare il quadro d'insieme nella direzione di elaborare ulteriormente la vostra idea

Concetto di utenza

- Nello sviluppo della vostra idea, pensate a quali **classi di utenza** potrebbero accedere al vostro servizio
- Non fermatevi al **consumer**... potrebbero anche esserci dei **producer**
- Valutate se possa esserci **ruolo attivo dell'utenza**, ovvero “possibilità di maggiore coinvolgimento”

Back e front-end

- Il servizio rappresenta il **back-end** della vostra soluzione e di fatto realizza l'aspetto centrale del vostro progetto d'esame
- Il **front-end**, web o mobile, è obbligatorio, ma non deve necessariamente essere avanzato (discrezione del gruppo)

Osservazioni:

- Il back-end ha un ruolo centrale, la UI ne rappresenta il dimostratore
- Il vostro progetto potrebbe implementare parte (gli aspetti più rilevanti) della vostra idea, non è obbligatorio che la realizzi per intero
- Il vostro servizio deve esporre un set consistente di funzionalità

Ruolo degli Open Data

Gli Open Data sono degli ingredienti posti sul tavolo

- **Domande:**
 - Q: Devo obbligatoriamente integrare alcuni Open Data?
 - A: No, solo se sono funzionali alla vostra idea
 - Q: Alcuni Open Data sono presenti in altre città, posso utilizzarli?
 - A: Certamente, se possono supportare la vostra idea
 - Q: Gli Open Data di cui avrei bisogno non esistono. Posso crearne di artificiali?
 - A: Certamente, se possono supportare la vostra idea. Potrebbe essere uno stimolo per il Comune per focalizzare attenzione in nuove direzioni.

Considerazioni sulla sicurezza



Valutare dove realisticamente serve

Valutazione attenta, confrontatevi anche al di fuori del gruppo (i.e., interviste amici/parenti)



Non solo scenari On/Off

Pensate ad Amazon. Dove interviene AuthN/AuthZ?



Barriera vs opportunità

Una registrazione o una fase di login, rappresentano una potenziale barriera all'utilizzo del servizio (di fatto sono step aggiuntivi)

Possono però abilitare servizi a valore aggiunto a scenari di business, servizi differenziati, etc.

Valutazione "costo"/"benefici"



Classi di utenza

Diverse classi di utenza potrebbero avere requisiti differenti

Valutazione delle vostre idee



**Rischio autoreferenzialità:
consultate persone esterne
(amici, parenti, etc.)**



**Raccogliete input,
empatizzate con
potenziali utenti, per
andare oltre l'esigenza
che avete intercettato**



**Integrate input,
l'intuizione che avete
avuto potrebbe essere
un'idea non matura**



**Continuate ad
elaborare l'idea anche
quando pensate di
averla già definita!**

Abbiamo provato a farlo con
un gruppo durante il
ricevimento, l'idea ha
ampliato il suo perimetro
iniziale

- Non esitate a chiedere
momenti di confronto!

Concetto e metodologia

Concetto

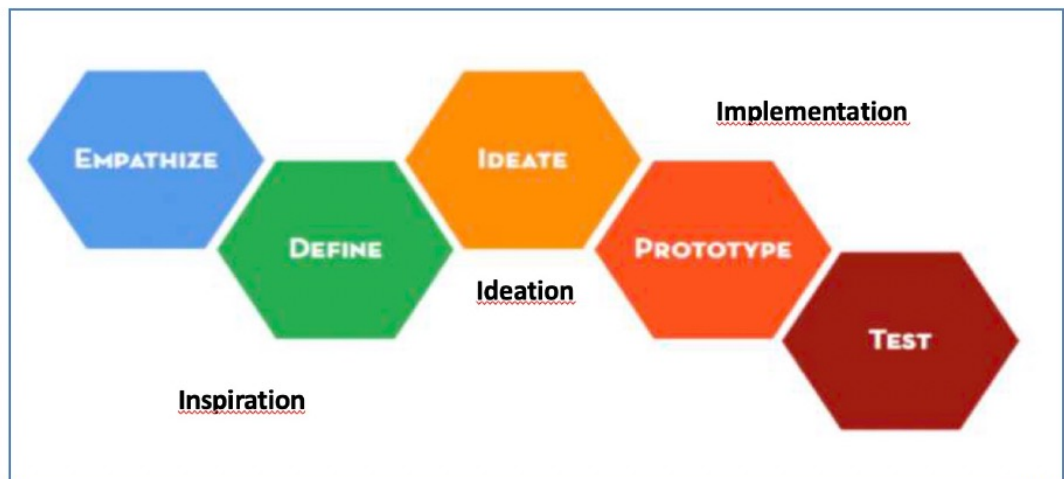


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La vostra **idea** deve necessariamente sottointendere il concetto di **servizio**

Metodologia



Workshop DT: Lunedì 22/09

Worksheet packet

Your NEW mission: **Design something useful and meaningful for your partner.**
Start by gaining **empathy**.

1 Interview

8min (2 sessions x 4 minutes each)

Notes from your first interview

d. @@@@

Switch roles & repeat Interview

2 Dig Deeper

6min (2 sessions x 3 minutes each)

Notes from your second interview



3 Capture findings 3min

Goals and Wishes:

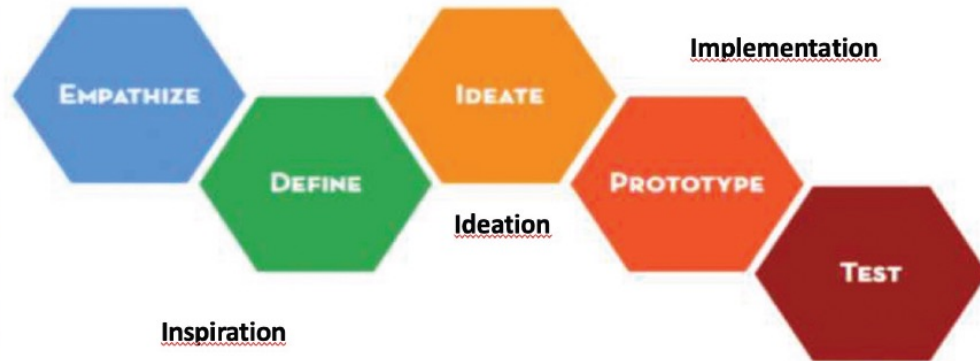
What is your partner trying to achieve?

*use verbs

Insights: New learnings about your partner's feelings and motivations. What's something you see about your partner's experience that maybe s/he doesn't see?"

*make inferences from what you heard

d. @@@@



partner's name/description

needs a way to _____ user's need

because (or "but ..." or "Surprisingly ...") _____ [circle one]

_____ insight



8 Build your solution.

Make something your partner can interact with!

[not here]

9 Share your solution and get feedback.

+ What worked...

- What could be improved...

? Questions...

! Ideas...



your user's needs. 5min

6 Share your solutions & capture feedback. 10min (2 sessions x 5 minutes each)

Notes

d. @@@@

Switch roles & repeat sharing.

Presentazione delle vostre idee (I)

Struttura del vostro elevator pitch

- Introduzione del **Team** (potrebbe essere la prima slide con nomi e titolo del vostro Progetto)
- Descrizione del **problema** o dell'**opportunità** che avete identificato
- Descrizione della vostra **soluzione**
- Descrizione del **valore** essa fornisce ed in che modo è diversa da altro o di valore unico

Workshop DT: Martedì 23/09

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Design Thinking

Methodology

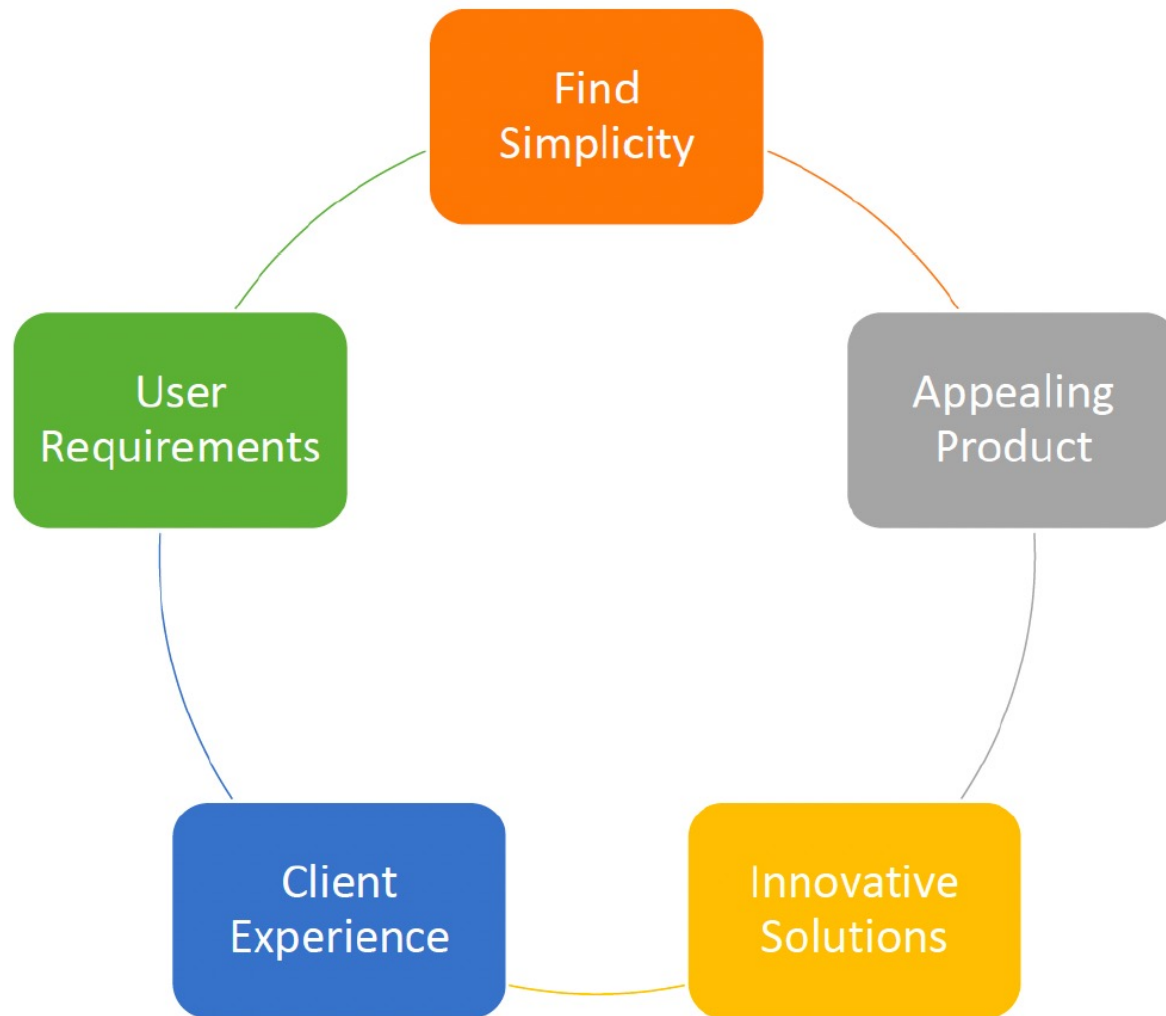
Design thinking

- Design thinking is a **methodology** that designers use to brainstorm and solve complex problems related to designing and design engineering
- It is also beneficial for designers to find **innovative, desirable and never-thought-before** solutions for **customers and clients**
- It is used extensively in the area of **healthcare and wellness, agriculture, food security, education**, financial services, and environmental sustainability, etc.

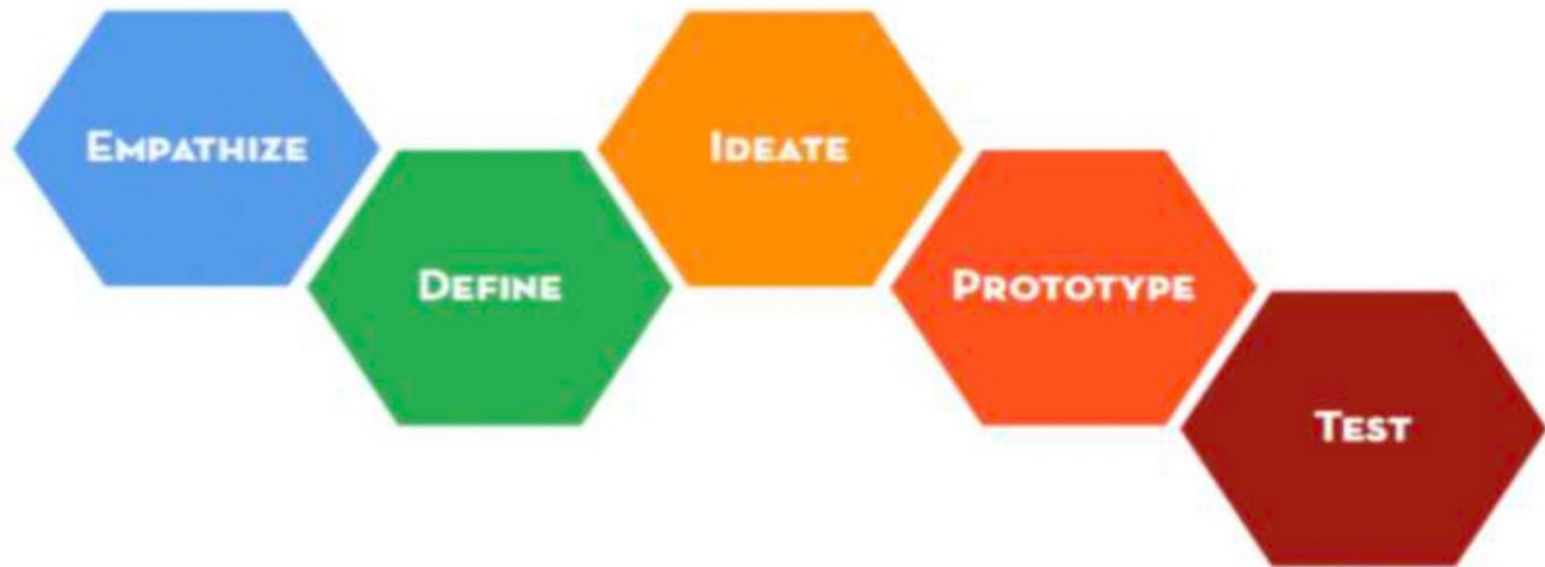
Use of Design Thinking

- Considered to be a **strategy for innovation**
- It is halfway between analytical thinking and intuitive thinking
- The **basic principle** of design thinking is that **innovation can be disciplined**
 - **Innovation** is a practice that can be **systematically approached** by a set of practical and meticulous **tools, methodologies, and frameworks**

Features of design thinking



The five-step process



- Design thinkers are expected **not to think of the following steps when working on one step**
 - For example it is not recommended to think of solutions, when the problem is being defined

Empathize Stage



- It involves **putting oneself into the shoes** of the customer or the **end-user** of our solution
 - We need to **understand the problems** faced by the **customer**
 - As design thinkers, we **need to empathize** with the **customer**
 - It involves the **process of analysis**
- **Empathy is the centerpiece of a human-centered design process**
- Observe what people do and how
 - It **gives feedback** about **what** they **think** and **feel**
 - Helps infer **intangible meaning** of those experiences
 - **uncover insights** → **give direction of innovative solutions**

Define Stage



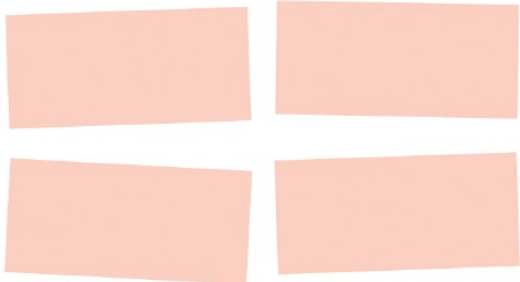
- It is time to define our problem → **problem statement**
- Information gathered → **coherent single statement**
- We have learnt the **problems** and **context** of our customers
 - User, needs, insights → **actionable statement**
- **Synthesis process** → bringing **clarity** and **focus** → **narrowing focus yields greater quantity/quality solutions**

Define Stage



“Framing the right problem is the only way to create the right solution”

HOW MIGHT WE..?

<i>What is the problem you want to solve?</i>	<i>Key Insights about the user</i> 
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HOW MIGHT WE HELP

Our user

SOLVE

The problem

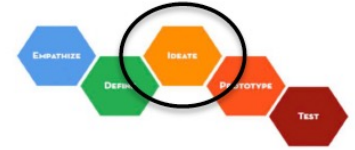
KNOWING THAT HE/SHE

?

needs/desires/insights we found out about the user

Picture from Design Thinking Playbook: <http://www.catalinacatana.com/wp-content/uploads/2018/10/Design-Thinking-playbook.pdf>

Ideate Stage



- At this stage a design thinker is supposed to bring to the table **as many ideas as possible**
- While brainstorming for ideas, it is not checked if the idea is **possible, feasible, viable** or not.
- Combine **rational thoughts** with **imagination**
- All solutions suggested by design thinkers are brought to the table and **thought over**
- Ideation techniques: **prototyping, mindmapping, bodystorming, sketching, ...**

Prototype Stage



This step deals with:

- building the ideas
- checking for their feasibility to arrive at the final solution.

In this step we take care of:

- Creation of experience (*for user, small scale, tangible*)
- Getting feedback (*from user*)
- Iteration (*through Empathize, Define, Ideate if necessary*)

End users comes into play

Prototype Stage



- Build the prototype → **iterative generation**
- **Don't waste too much of time/money** on building a single prototype
- The prototypes must be **built for and with the end user in mind**
- The prototype must create **an experience for the user**
- Think of **open questions** that the user can shoot towards you when he experiences the prototype
- There is no **value in the prototype** in case the **user** does not feel comfortable and satisfied with it

“Build to think and test to learn”

Test Stage



- The **idea** that seems the **best** according to the feedback of the customers and end users in the prototype phase will be executed
- **Full scale testing** to understand what actually works and what does not.
- This step can be:
 - the **most rewarding**, if the prototypes succeed to give positive results,
 - the **most annoying**, if the prototype fails.
- After testing
 - the entire process of design thinking **may have to be repeated**.
 - If the end user approves the solution, then the process of design thinking **stops here**.

Test Stage

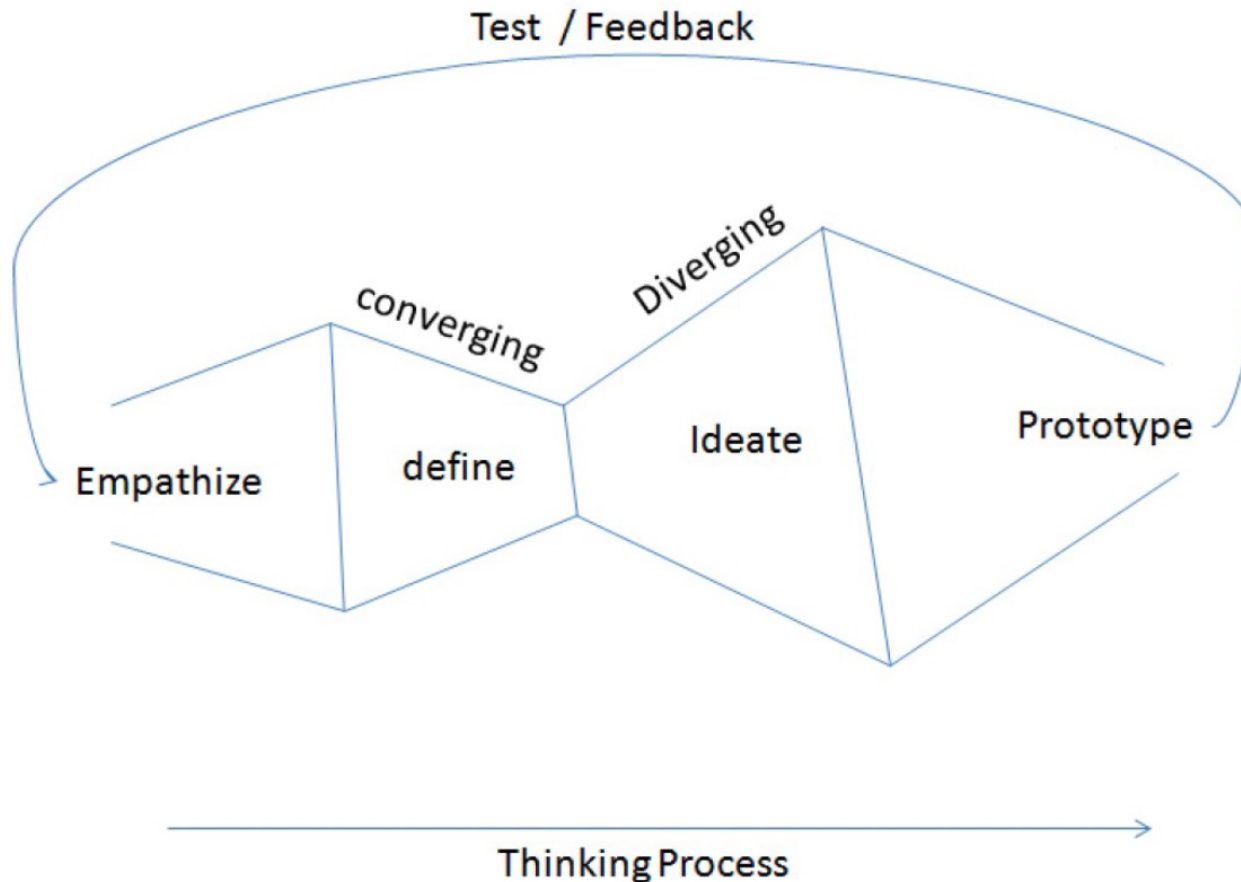


- Why test
 - refine prototype and solutions
 - learn more about users
 - refine problem statement
- How to test:
 - Show don't tell
 - Create experiences
 - Ask users to compare

“Testing is an opportunity to learn about your solution and your user”

Iterative process

Putting everything together



Design thinking process and IDEO shopping cart example

Design thinking process

[https://www.youtube.com/watch?v= r0VX-aU T8](https://www.youtube.com/watch?v=r0VX-aU_T8)

Take a few minutes to watch the following video about the IDEO Shopping Cart:

<https://www.youtube.com/watch?v=M66ZU2PClCM>

Reframe the problem.

3 Capture findings 3min

Goals and Wishes:

What is your partner trying to achieve?

*use verbs

Insights: New learnings about your partner's feelings and motivations. What's something you see about your partner's experience that maybe s/he doesn't see?*

*make inferences from what you heard

4 Take a stand with a point-of-view 3min



partner's name/description

needs a way to

user's need

because (or "but..." or "Surprisingly...")

[circle one]

insight

Ideate: generate alternatives to test.

5 Sketch at least 5 *radical* ways to meet your user's needs. 5min



write your problem statement above

A large, empty rectangular box with rounded corners, intended for sketching a radical solution.A large, empty rectangular box with rounded corners, intended for sketching a radical solution.A large, empty rectangular box with rounded corners, intended for sketching a radical solution.A large, empty rectangular box with rounded corners, intended for sketching a radical solution.A large, empty rectangular box with rounded corners, intended for sketching a radical solution.

6 Share your solutions & capture feedback. 10min (2 sessions x 5 minutes each)

Notes

A large, empty rectangular box with rounded corners, intended for taking notes during the sharing and feedback session.

Iterate based on feedback.

7 Reflect & generate a new solution. 3min

Sketch your big idea, note details if necessary!

Build and test.

8 Build your solution.

Make something your partner can interact with!

[not here]

7min

d. 

9 Share your solution and get feedback.

+ What worked...

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An Introduction to Design Thinking

"Wallet" Edition

Facilitator's Guide:
Script, talking points, takeaways,
and setup considerations inside.



The Wallet Project

A little background on the project

The project you're holding in your hands is an iteration on the d.school's iconic "Wallet Project." The original wallet project was created as an introduction to design thinking for the d.school's inaugural Boot Camp class in the Winter of 2006. It has since been contributed to, modified, stretched, and evolved by many d.school collaborators.

The Wallet Project is an immersive activity meant to give participants a full cycle through the design thinking process in as short a time as possible. The project itself gives facilitators the opportunity to touch on the fundamental values of the d.school—human-centered design, a bias towards action, and a culture of iteration and rapid prototyping—without attempting to communicate all of the methods and activities that the term "design thinking" encompasses.

Why did we choose a wallet as the starting point for the introductory design challenge?

- * everyone has experience with a wallet, or another way to carry cash, cards, and ID
- * the wallet and its contents have the potential to evoke a range of meaning and the larger context of a person's life
- * wallet as a starting point enables a wide range of potential innovation outcomes (we've seen objects, experiences, services, systems, and spaces!)
- * having a physical artifact in-hand allows for immediate recall of experiences (participants can gain empathy for one another in the room)

It is certainly possible to facilitate a similar project with a different topic—and you may choose to in order to amp up a particular aspect you think is important for your group of participants. For example, we have done the "oral-hygiene project" to make it more personal and had participants observe their partners in their homes previous to the beginning of the workshop. We also do a "redesign the gift-giving experience" version, with the aim to encourage participants to create services, experiences, and systems. If you decide to change the topic [you can simply change the instructions for the first two steps], be mindful of the considerations we listed above and scope it in a way that is both wide-open to possibilities but also manageable for participants.

Why project-based, team-facilitated learning?

Having created learning experiences for students of all cultures, ages, disciplines, and industries, we have found that engaging in projects provides a much richer learning experience than listening to a "talking head" does. As such, our bias is to provide limited scaffolding to allow participants to do, and then to facilitate a reflection that invites the participants themselves to extract the meaningful learning opportunities from the experience. We teach in teams at the d.school because we have found that this approach tends to create a conversation in the classroom, as opposed to the one-way communication that often transpires in more lecture-driven formats. We relish the diversity of perspectives that emerge when faculty from diverse backgrounds instruct, and even disagree in front of, a class. One great way to run the Wallet Project with two people, is for one to take *lead* (concentrating on communicating the instructions, logistics, and timing), and the other person to provide *color* (communicating the nuances, offering encouragement, and providing helpful tips).

Your NEW mission: **Design something useful and meaningful for your partner.**
Start by gaining empathy.

1 Interview

8min (2 sessions x 4 minutes each)

Notes from your first interview

d.

Switch roles & repeat Interview

2 Dig Deeper

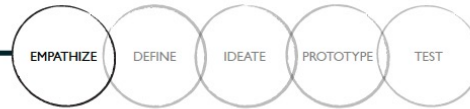
6min (2 sessions x 3 minutes each)

Notes from your second interview

Switch roles & repeat Interview

Start by gaining empathy.

How to facilitate these steps



1—Interview your Partner

Have participants partner up in pairs.

It helps to refer to “Partner A” and “Partner B” to simplify your language in these interactive steps.

“Your challenge is to design something useful and meaningful to your partner.”

“The most important part of designing for someone is to gain empathy for that person.”

“One way to do this is to have a good conversation.”

Be clear about the logistics of the interviews:

“Partner A will have four minutes to interview Partner B, and then we will tell you when to switch.”

“As a starting point, ask your partner to walk you through the contents of their wallet.”

“When do they carry their wallet? Why do they have a particular card in there? What do the things in their wallet tell you about their life?”

Tell them to take note of things they find interesting or surprising.

“Let’s begin!” (Don’t forget to start playing the upbeat music now.)

2—Dig deeper

After the first set of interviews, tell them to follow up on things that intrigued them during the first interview.

“Try to dig for stories, feelings, and emotion.”

“Ask ‘WHY?’ often”

“Forget about the wallet, find out what’s important to your partner.”

“Why does he still carry a picture of his ex-girlfriend? When was a time he carried a lot of cash? What does she remember most about her first paying job?”

Remind them you will let them know when time is up.

...“Time to switch! Again, make note of any unexpected discoveries along the way, capture quotes!”

Your NEW mission
Design something useful and meaningful for your partner.
Start by gaining empathy.

1 Interview

8min (2 sessions x 4 minutes each)

Notes from your first interview

1 2 3 4 5 6

Switch roles & repeat interview

2 Dig Deeper

6min (2 sessions x 3 minutes each)

Notes from your second interview

Switch roles & repeat interview

Reframe the problem.

3 Capture findings 3min

Goals and Wishes:

What is your partner trying to achieve?

*use verbs

Insights: New learnings about your partner's feelings and motivations. What's something you see about your partner's experience that maybe s/he doesn't see?*

*make inferences from what you heard

4 Take a stand with a point-of-view 3min



partner's name/description

needs a way to

user's need

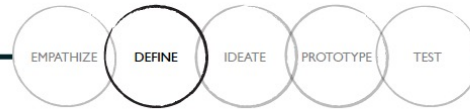
because (or “but . . .” or “Surprisingly . . .”)

[circle one]

insight

Reframe the problem.

How to facilitate these steps



3—Capture findings

Tell them to *individually* take three minutes to collect their thoughts and reflect on what they've learned about their partner.

“Synthesize your learning into two groups: your partner’s goals and wishes, and insights you discovered.”

“Use verbs to express the goals and wishes.”

“These are his needs related to his wallet and life. Think about both physical and emotional needs.”

“For example maybe your partner *needs* to minimize the number of things he carries . . . or needs to feel like she is supporting the local community and economy.”

“‘Insights’ are discoveries that you might be able to leverage when creating solutions.”

“For example, you might have discovered the insight that *buying with cash makes your partner value the purchase* more and take more care with decisions.”

“Or, that she sees a wallet as a reminder and organizing system, not a carrying device.”

4—Take a stand with a point-of-view

Now tell them to select the most compelling need and most interesting insight to articulate a point-of-view.

“This is your point of view.”

“Take a stand by specifically stating the meaningful challenge you are going to take on.”

“This is the statement that you’re going to address with your design, so make sure it’s juicy and actionable!”

“It should feel like a problem worth tackling!”

“Your point-of-view might be:

‘Janice need a way to feel she has access to all her stuff and is ready to act. Surprisingly, carrying her purse makes her feel less ready to act, not more.’

Or ‘Arthur needs a way to socialize with his friends while eating healthy, but he feels like he isn’t participating if he isn’t holding a drink.’”

Reframe the problem.

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d.○○○○○

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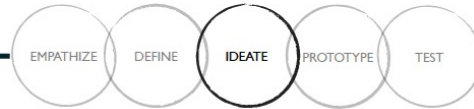
6 Share your solutions & capture feedback. 10min (2 sessions x 5 minutes each)

Notes

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Ideate: generate alternatives to test.

How to facilitate these steps



5—Sketch to Ideate

Have them rewrite their problem statement at the top of the page.

Remind them they are now creating solutions to the new challenge they've identified.

They then to sketch a lot of ideas, and to try to create a number of different ideas.

“GO FOR VOLUME!”

“This is time for idea generation, not evaluation—you can evaluate your ideas later.”

You can even suggest a friendly competition to come up with the most ideas --

“See if you can come up with at least 7 ideas!”

Remind them they are not necessarily designing a wallet; instead they should create solutions to problem statement they just created.

“Remember to be VISUAL—use words just when necessary to call out details.”

... “One minute left! Try to sketch at least 2 more wildly different ways to address your problem statement!”

6—Share solutions and capture feedback

“Now it's time to share your sketches with your partner!”

“Partner A, share your sketches with Partner B first, and then we will tell you to switch after four minutes.”

Tell them to note likes/dislikes and builds on the idea, but also listen for new insights.

“Spend the time listening to your partners reactions and questions.”

“This is not just about validating your ideas.”

“Fight the urge to explain and defend your ideas.”

“This is another opportunity to learn more about your partner's feelings and motivations.”

Tell them to switch after time is up for the first session.

Ideate: generate alternatives to test.

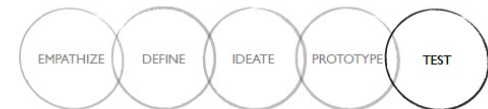
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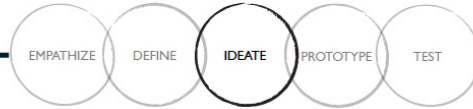
Iterate based on feedback.

7 Reflect & generate a new solution. 3min

Sketch your big idea, note details if necessary!

Iterate based on feedback.

How to facilitate this step



7—Reflect & generate a new solution

They've had a chance to share their sketches with their partners and collect feedback.

“Now, take a moment to consider what you have learned both about your partner, **and** about the solutions you generated.”

“From this new understanding of your partner and his or her needs, sketch a new idea.”

Tell them this solution may be a variation on an idea from before or something completely new.

They are still addressing a problem statement that *they* articulated, but you might point out that their previous problem statement may need to change to incorporate the new insights and needs they discovered.

“Try to provide as much detail and color around your idea as possible.”

“How might this solution fit into the context of your partner's life?”

“When and how might they handle or encounter your solution?”

While participants are working, grab the prototyping materials if you have not already.

Iterate based on feedback.

7 Reflect & generate a new solution. 3min

Sketch your big idea, note details if necessary!

d ○○○○○

Build and test.

8 Build your solution.

Make something your partner can interact with!

[not here]

7min

d. 

9 Share your solution and get feedback.

✚ What worked...

▣ What could be improved...

? Questions...

! Ideas...

8min (2 sessions x 4 minutes each)

Build and test.

How to facilitate these steps

8—Build!

Tell participants to use the idea they just sketched as a blueprint for a tangible manifestation of their solution.

“Create a physical prototype of your solution.”

Explain that they should not simply make a scale model of their idea to explain the idea.

They should create an experience that their partner can react to.

They could decide to test just one aspect of the overall solution.

“MAKE something that your partner can engage and interact with.”

“If your solution is a service or a system,

create a scenario that allows your partner to experience this innovation.”

“Use whatever materials are available to you—including space!”

Provide some urgency and excitement in your voice—you want to get them building immediately.

“Be scrappy and quick—you only have a few minutes!!”

9—Share your solution and get feedback

Explain that one partner will have time to share and collect feedback, and then they will switch so the other partner can share.

“Now you’re going to have the opportunity to share your prototype with your partner.”

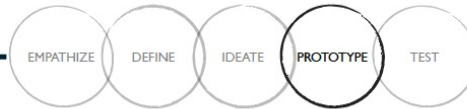
Validation of the prototype is not the point—it should be an artifact that facilitates a new, targeted conversation.

“When you test, LET GO of your prototype, physically and emotionally.”

“Your prototype is NOT PRECIOUS, but the feedback and new insights it draws out are!”

“Don’t defend your prototype; instead, watch how your partner uses and misuses it.”

Tell them to jot down things their partner liked and didn’t like about the idea, as well as questions that emerged and new ideas that came up.



Build and test.

8 Build your solution.

Make something your partner can interact with!

(not here)

7min

⌚ ⌚ ⌚ ⌚ ⌚ ⌚

9 Share your solution and get feedback.

➤ What worked...

➤ What could be improved...

? Questions...

! Ideas...

8min (2 sessions x 4 minutes each)

